CREATING A DATABASE AND TABLE:

Interactive SQL Course

Input

Run SQL

```
CREATE TABLE books(
    book_id INTEGER PRIMARY KEY AUTOINCREMENT,
    title VARCHAR(255) NOT NULL,
    isbn VARCHAR(55) NOT NULL UNIQUE,
    published_year INT,
    CHECK(published_year < 2027)
);

CREATE TABLE members(
    member_id INTEGER PRIMARY KEY AUTOINCREMENT,
    full_name VARCHAR(100) NOT NULL,
    email VARCHAR(150) NOT NULL UNIQUE
);
```

Output

book_id	title	isbn	published_year
1	The Alchemist	9780061122415	1988
2	Clean Code	9780132350884	2008
3	Atomic Habits	9780735211292	2018

member_id	full_name	email
1	Nuguri Siri	siripanchajanya@gmail.com
2	Spandana	s.spandana@klh.edu.in
3	Bhargavi	kvbhargavi@klh.edu.in

Available Tables

Books

book_id	title	isbn	published_year
1	The Alchemist	9780061122415	1988
2	Clean Code	9780132350884	2008
3	Atomic Habits	9780735211292	2018

Customers

customer_id	first_name	last_name	age	country
1	John	Doe	31	USA
2	Robert	Luna	22	USA
3	David	Robinson	22	UK
4	John	Reinhardt	25	UK
5	Betty	Doe	28	UAE

Members

member_id	full_name	email

INSERTING AND CREATING TABLE AND SELECTING

Interactive SQL Course

Input

Run SQL

```
SELECT * FROM books;
SELECT * FROM members;


CREATE TABLE student_info(
    student_id INTEGER PRIMARY KEY AUTOINCREMENT,
    full_name VARCHAR(2000) NOT NULL,
    email VARCHAR(2000) NOT NULL UNIQUE
);

INSERT INTO student_info(full_name, email) VALUES
('Nuguri Siri', 'siripanchajanya@gmail.com'),
('abcdefghijk', 'abc@klh.edu.in');
```

Output

book_id	title	isbn	published_year
1	The Alchemist	9780061122415	1988
2	Clean Code	9780132350884	2008
3	Atomic Habits	9780735211292	2018

member_id	full_name	email
1	Nuguri Siri	siripanchajanya@gmail.com
2	Spandana	s.spandana@klh.edu.in
3	Bhargavi	kvbhargavi@klh.edu.in

Available Tables

Books

book_id	title	isbn	published_year
1	The Alchemist	9780061122415	1988
2	Clean Code	9780132350884	2008
3	Atomic Habits	9780735211292	2018

Customers

customer_id	first_name	last_name	age	country
1	John	Doe	31	USA
2	Robert	Luna	22	USA
3	David	Robinson	22	UK
4	John	Reinhardt	25	UK
5	Betty	Doe	28	UAE

Members

member_id	full_name	email

SELECTING FROM TABLE:

Interactive SQL Course

Input

Run SQL

```
SELECT COUNT(*) AS total_students
FROM student_marks1;

SELECT SUM(marks) AS total_marks
FROM student_marks1;

SELECT AVG(marks) AS average_marks
FROM student_marks1;

SELECT MAX(marks) AS maximum_marks
FROM student_marks1;

SELECT MIN(marks) AS minimum_marks
```

Output

book_id	title	isbn	published_year
1	The Alchemist	9780061122415	1988
2	Clean Code	9780132350884	2008
3	Atomic Habits	9780735211292	2018

member_id	full_name	email
1	Nuguri Siri	siripanchajanya@gmail.com
2	Spandana	s.spandana@klh.edu.in
3	Bhargavi	kvbhargavi@klh.edu.in

Available Tables

Books

book_id	title	isbn	published_year
1	The Alchemist	9780061122415	1988
2	Clean Code	9780132350884	2008
3	Atomic Habits	9780735211292	2018

Customers

customer_id	first_name	last_name	age	country
1	John	Doe	31	USA
2	Robert	Luna	22	USA
3	David	Robinson	22	UK
4	John	Reinhardt	25	UK
5	Betty	Doe	28	UAE

Members

member_id	full_name	email

OUTPUTS:

```
-- GROUP BY

SELECT subject, SUM(marks) AS total_marks
FROM student_marks1
GROUP BY subject;

SELECT subject, AVG(marks) AS average_marks
FROM student_marks1
GROUP BY subject;

SELECT subject, COUNT(*) AS num_students
FROM student_marks1
GROUP BY subject;
```

Output

aws	98.5
aws1	99.9
azure	82.25
dbms	95.75
english	97.4

subject	num_students
Math	5
aws	1
aws1	1
azure	1
dbms	1

Customers

customer_id	first_name	last_name	age	country
1	John	Doe	31	USA
2	Robert	Luna	22	USA
3	David	Robinson	22	UK
4	John	Reinhardt	25	UK
5	Betty	Doe	28	UAE

Orders

order_id	item	amount	customer_id
1	Keyboard	400	4
2	Mouse	300	4
3	Monitor	12000	3
4	Keyboard	400	1
5	Mousepad	250	2

Shippings

shipping_id	status	customer
1	Pending	2
2	Pending	4
3	Delivered	3

```sql
-- GROUP BY

SELECT subject, SUM(marks) AS total_marks
FROM student_marks1
GROUP BY subject;

SELECT subject, AVG(marks) AS average_marks
FROM student_marks1
GROUP BY subject;

SELECT subject, COUNT(*) AS num_students
FROM student_marks1
GROUP BY subject;
```

Output

11	Ravi	Math	85.5
12	Sita	Math	92.75
14	Priya	Math	88.9
16	subbusir	aws	98.5
17	dwaraka	dbms	95.75
18	ranjani	english	97.4
19	kavithamam	awsl	99.9

roll_no	name	subject	marks
12	Sita	Math	92.75
16	subbusir	aws	98.5
17	dwaraka	dbms	95.75
18	ranjani	english	97.4

Customers

customer_id	first_name	last_name	age	country
1	John	Doe	31	USA
2	Robert	Luna	22	USA
3	David	Robinson	22	UK
4	John	Reinhardt	25	UK
5	Betty	Doe	28	UAE

Orders

order_id	item	amount	customer_id
1	Keyboard	400	4
2	Mouse	300	4
3	Monitor	12000	3
4	Keyboard	400	1
5	Mousepad	250	2

Shippings

shipping_id	status	
1	Pending	
2	Pending	4
3	Delivered	3


```sql
-- GROUP BY

SELECT subject, SUM(marks) AS total_marks
FROM student_marks1
GROUP BY subject;

SELECT subject, AVG(marks) AS average_marks
FROM student_marks1
GROUP BY subject;

SELECT subject, COUNT(*) AS num_students
FROM student_marks1
GROUP BY subject;
```

Output

11	Ravi	Math	85.5
12	Sita	Math	92.75
14	Priya	Math	88.9
16	subbusir	aws	98.5
17	dwaraka	dbms	95.75
18	ranjani	english	97.4
19	kavithamam	awsl	99.9

roll_no	name	subject	marks
12	Sita	Math	92.75
16	subbusir	aws	98.5
17	dwaraka	dbms	95.75
18	ranjani	english	97.4

Customers

customer_id	first_name	last_name	age	country
1	John	Doe	31	USA
2	Robert	Luna	22	USA
3	David	Robinson	22	UK
4	John	Reinhardt	25	UK
5	Betty	Doe	28	UAE

Orders

order_id	item	amount	customer_id
1	Keyboard	400	4
2	Mouse	300	4
3	Monitor	12000	3
4	Keyboard	400	1
5	Mousepad	250	2

Shippings

shipping_id	status	
1	Pending	
2	Pending	4
3	Delivered	3

```sql
-- GROUP BY

SELECT subject, SUM(marks) AS total_marks
FROM student_marks1
GROUP BY subject;

SELECT subject, AVG(marks) AS average_marks
FROM student_marks1
GROUP BY subject;

SELECT subject, COUNT(*) AS num_students
FROM student_marks1
GROUP BY subject;
```

Output

average_marks
89.96

maximum_marks
99.9

minimum_marks
78.4

roll_no	name	subject	marks
11	Ravi	Math	85.5
12	Sita	Math	92.75
14	Priya	Math	88.9
16	subbusir	aws	98.5

Customers

customer_id	first_name	last_name	age	country
1	John	Doe	31	USA
2	Robert	Luna	22	USA
3	David	Robinson	22	UK
4	John	Reinhardt	25	UK
5	Betty	Doe	28	UAE

Orders

order_id	item	amount	customer_id
1	Keyboard	400	4
2	Mouse	300	4
3	Monitor	12000	3
4	Keyboard	400	1
5	Mousepad	250	2

Shippings

shipping_id	status	
1	Pending	2
2	Pending	4
3	Delivered	3


```sql
-- GROUP BY

SELECT subject, SUM(marks) AS total_marks
FROM student_marks1
GROUP BY subject;

SELECT subject, AVG(marks) AS average_marks
FROM student_marks1
GROUP BY subject;

SELECT subject, COUNT(*) AS num_students
FROM student_marks1
GROUP BY subject;
```


Output

roll_no	name	subject	marks
11	Ravi	Math	85.5
12	Sita	Math	92.75
13	Anil	Math	78.4
14	Priya	Math	88.9
15	Vijay	Math	80.25
16	subbusir	aws	98.5
17	dwaraka	dbms	95.75
18	ranjani	english	97.4
19	kavithamam	awsl	99.9
20	seetha	azure	82.25

Customers

customer_id	first_name	last_name	age	country
1	John	Doe	31	USA
2	Robert	Luna	22	USA
3	David	Robinson	22	UK
4	John	Reinhardt	25	UK
5	Betty	Doe	28	UAE

Orders

order_id	item	amount	customer_id
1	Keyboard	400	4
2	Mouse	300	4
3	Monitor	12000	3
4	Keyboard	400	1
5	Mousepad	250	2

Shippings

shipping_id	status	customer
1	Pending	2
2	Pending	4
3	Delivered	3